



Quant Week 7

3/27/25



Agenda



1. Icebreaker Question
2. Weekly Sequence (Hard)
3. Brain Teaser (Easy)
4. Project



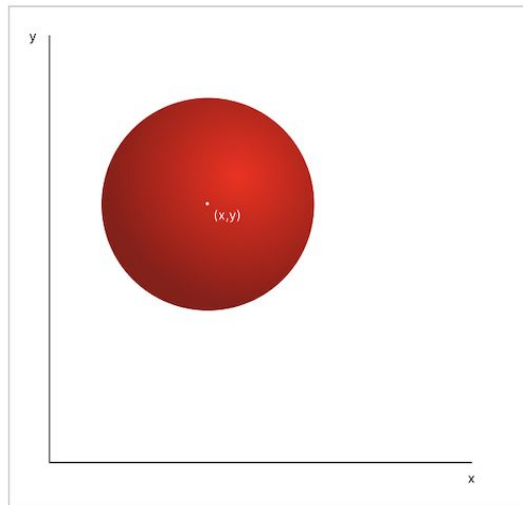
Ice Breaker

What did you eat today?

How much protein was in your meal?



Brain Teaser



Completing the Square Method mathgymath

Step 1 Rearrange if Necessary *Move the constants to one side	Step 2 $+ (b/2)^2$ to both sides	Step 3 Factor & Solve
$x^2 - 6x + 9 = 25$ $x^2 - 6x = 16$	$(\frac{-6}{2})^2 = (-3)^2 = 9$ $x^2 - 6x + 9 = 16 + 9$ $x^2 - 6x + 9 = 25$	$x^2 - 6x + 9 = 25$ $(x-3)^2 = 25$ $x-3 = \pm 5$ $x = 3 \pm 5$ $x = 8$ $x = -2$

Consider the following equation:

$$x^2 + y^2 - 16x - 10y + 73 = 0,$$

which represents a circle in Cartesian coordinates. Based on this provided equation, determine the x-coordinate of the center of the circle.

Please fill in your answer as an integer.



Weekly Sequence

3, 3, 6, 12, 60, 660, ?



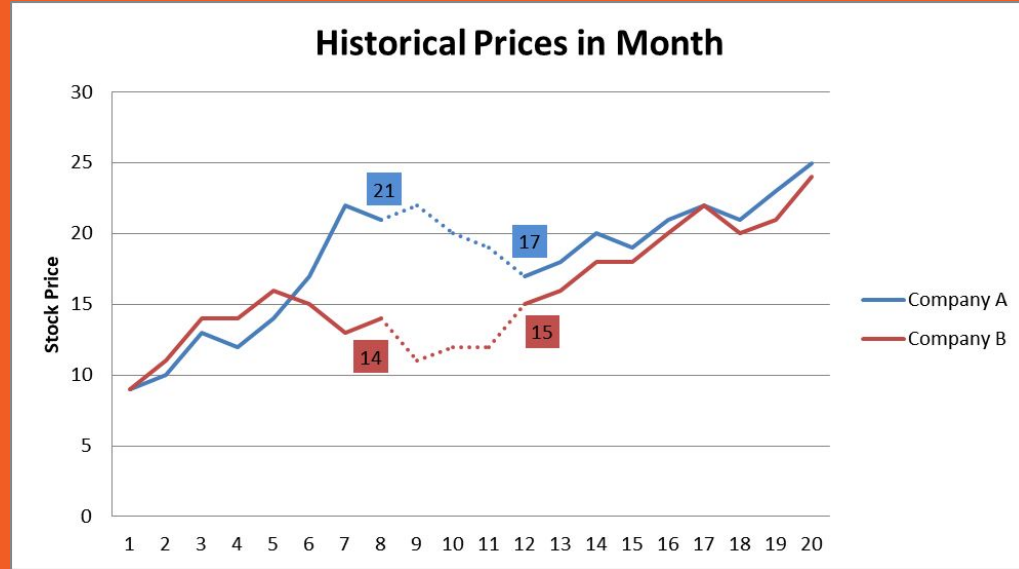
Some Trading Strategies

Pairs Trading

This strategy assumes that price changes in stocks in similar industries will tend to move together.

Holding all else equal, if Ford stock moves up, GM stock should move up as well

If Ford moves up and GM does not, it would be a good time to buy GM or short Ford.



Some Trading Strategies

RSI Index

$$RSI_{\text{step one}} = 100 - \left[\frac{100}{1 + \frac{\text{Average gain}}{\text{Average loss}}} \right]$$

RSI indicates if a stock is overbought or oversold.

It does this by keeping a moving average over a period of time of the average gain and average loss.

Values will be between 0 and 100, values over 70 indicate the stock is overbought and values under 30 indicate the stock is oversold.

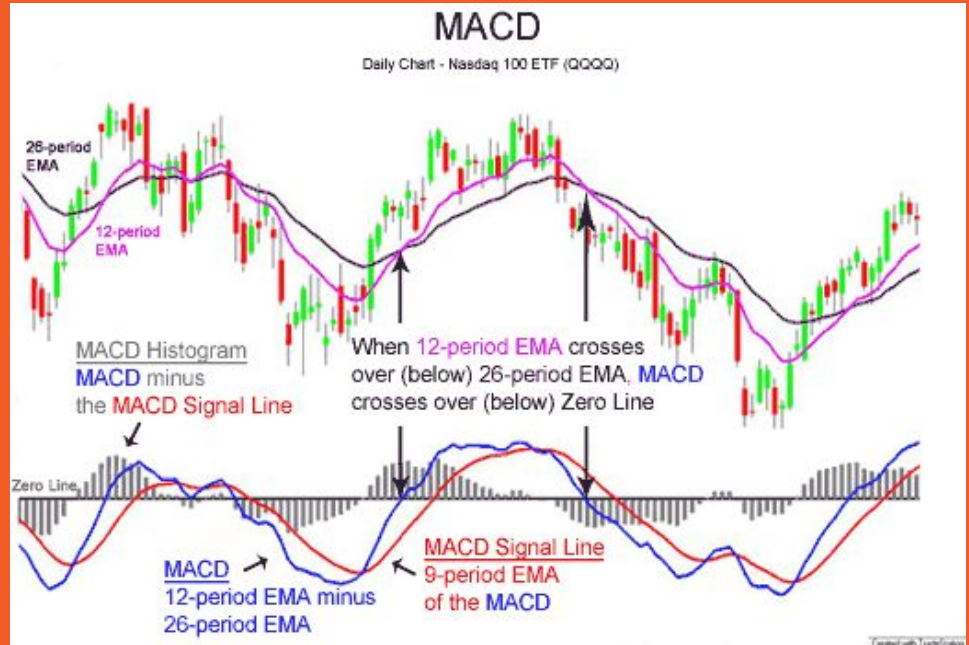
Some Trading Strategies

MACD

Moving Average
Convergence/Divergence

This indicator graphs the moving average over two different time periods.

When the lines cross it shows changing strength/momentum and can indicate to buy or sell



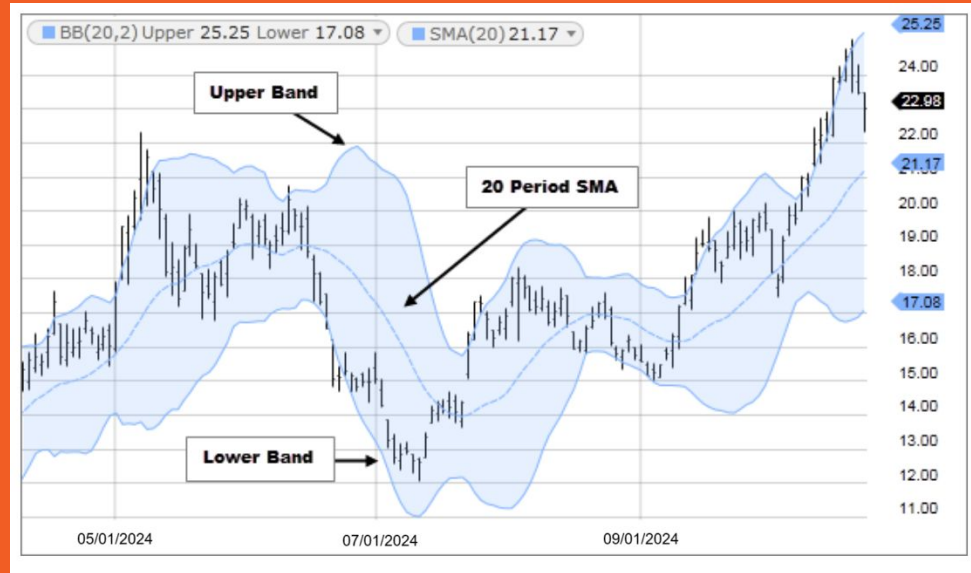
Some Trading Strategies

Bollinger Bands

Keep a moving average of the price of the stock.

The upper and lower bands are one standard deviation away from the current stock price.

If the price breaks through this level it indicates that the price is either high or low compared to the stock's true value.



Some Trading Strategies

Swing Trading

Attempting to buy and sell within a short time period and take advantage of natural price fluctuations.

Subreddit r/swingtrading has 100k people talking about different ways to implement this



Project Discussion/Work Time

